

Varying effects of climate change on the photosynthesis and calcification of crustose coralline algae: Implications for settlement of coral larvae

Jennifer M. Sneed^{a,*}, Justin E. Campbell^{a,b}, Audrey Looby^{a,c,d}, Aurora Giorgi^e, Valerie J. Paul^a

^a Smithsonian Marine Station at Fort Pierce, 701 Seaway Dr., Ft. Pierce, FL 34949 USA

^b Department of Biological Sciences and the Center for Coastal Oceans Research, Institute of Environment, Florida International University, North Miami, FL 33181 USA

^c Nature Coast Biological Station, Institute of Food and Agricultural Sciences, University of Florida, Cedar Key, FL, 32625 USA

^d Fisheries and Aquatic Sciences, Institute of Food and Agricultural Sciences, University of Florida, Gainesville, FL, 32611 USA

^e Red Sea Research Center, King Abdullah University of Science and Technology, Thuwal, 23955 Saudi Arabia

ARTICLE INFO

Keywords:

Larval settlement
Ocean warming
Ocean acidification
Hydrolithon boergeri
Coral reef ecology
Paragoniolithon solubile

ABSTRACT

Coral recruitment is critical to the maintenance of healthy coral reef ecosystems. Many coral species settle preferentially on certain crustose coralline algae (CCA) (e.g., *Hydrolithon boergeri*) over others (e.g., *Paragoniolithon solubile*). Calcifying organisms like CCA are particularly susceptible to ocean acidification (OA), and settlement behavior of larvae may be compromised as seawater temperatures increase (ocean warming; OW) and pH levels decrease as a result of climate change. Here, we examine the effects of future seawater conditions (OW and OA) on the calcification and photosynthetic efficiency of two CCA species, *H. boergeri* and *Pa. solubile*. We also examine the effects of conditioning CCA in combined OA and OW on the settlement preferences of three coral species, *Acropora palmata*, *A. cervicornis* and *Porites astreoides*. *Acropora palmata* and *Pa. astreoides* demonstrated a preference for *H. boergeri* over *Pa. solubile* in choice experiments after short-term treatment (7–21 days) and this preference was not affected by future seawater conditions. *A. cervicornis* did not demonstrate a CCA preference under any treatment. *Pa. astreoides* did not demonstrate a CCA preference in no-choice assays and settlement was unaffected by OW and OA even after the longest exposure (99 days). Both CCA had reduced photosynthetic efficiency after exposure to future seawater conditions. However, net calcification rate was reduced in *H. boergeri* but not *Pa. solubile* after exposure to future seawater conditions. These results demonstrate that while climate change may differentially affect the physiological functioning of various species of CCA, coral settlement preferences are unlikely to be altered.

1. Introduction

Coral reef degradation is occurring rapidly on a global scale [1–3] as reefs face both local (e.g., pollution, overfishing) and global threats (i.e., climate change). As disturbances occur in increasingly rapid succession, reefs lack the time necessary to recover and are therefore declining [4, 5]. Healthy coral reefs are often characterized by a high cover of both crustose coralline algae (CCA) and reef-building corals [6] and the abundance of these two types of organisms can be interdependent. In healthy coral reef ecosystems, CCA aid in reef accretion and stabilization by cementing loose rubble and sediment [7]. This function can be especially critical in areas exposed to disturbances that cause coral mortality and/or fragmentation of carbonate reef structures. CCA also

provide hard substrate conducive to successful coral recruitment and emit cues that induce coral larval settlement [8–11]. In addition to the direct relationships between CCA and corals, CCA also create habitat for other organisms and are important settlement substrata for a number of other marine invertebrates [7].

While reefs experience varying levels of local stressors, predominantly based on the degree of anthropogenic influence, changes in seawater temperature and pH caused by climate change affect reefs globally. According to the 2021 report from the Intergovernmental Panel on Climate Change (IPCC), by the year 2100, global oceanic pH could decrease by about 0.3 units and sea surface temperatures (SST) could increase by as much as 4.0 °C [12]. CCA are calcifying organisms considered to be among the most susceptible to ocean acidification (OA)

* Corresponding author.

E-mail address: SneedJ@si.edu (J.M. Sneed).

<https://doi.org/10.1016/j.ecochg.2024.100090>

Received 28 February 2024; Received in revised form 18 November 2024; Accepted 23 November 2024

Available online 1 December 2024

2666-9005/Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

as they secrete high Mg-calcite which is the most soluble polymorph of CaCO_3 [13–15]. In mesocosm experiments and along natural, in-situ gradients, CCA abundance declines and community composition is significantly altered as pCO_2 levels increase [16–20]. Because calcification is sensitive to changes in seawater carbonate chemistry, many studies have focused on the effects of increased pCO_2 on CCA calcification. Most of these studies indicate that CCA are likely to undergo reduced calcification under projected 21st century climate scenarios, resulting in net dissolution in some cases [21–25]. This response appears to be variable among species, experimental conditions and local environmental conditions [26–28], and there is recent evidence that CCA may be able to acclimate to increased pCO_2 or even adapt to these conditions over multiple generations [29,30]. For some calcifying algae, ocean warming (OW) may mediate the negative effect of OA on growth and calcification by increasing metabolism and photosynthesis [31,32]. However, the few studies focusing on the combined effects of OA and OW on tropical CCA species have demonstrated mixed results, with some finding synergistically negative effects of the two stressors [33] and others finding pCO_2 as the driving factor with little influence of increased temperature [26].

Changes in ocean pH and temperature have negative effects on coral recruitment through both direct effects on the coral itself and indirect effects on the settlement substrata [34–37]. As seawater becomes warmer and more acidic, declines in CCA abundance will likely contribute to reef degradation by destabilizing the substrate and reducing appropriate habitat and cues for coral recruitment, further exacerbating coral declines [38]. The extent of the impacts on coral settlement will be dependent, in part, on the susceptibility of certain CCA species to climate change. Several coral species have demonstrated preferences for CCA species in the genera *Hydrolithon* and/or *Titanoderma* over other co-occurring species [9–11,39], but it is unknown whether these preferences will be retained under future climate conditions. The majority of current research on CCA responses to climate change has focused on relatively few species and largely neglects *Hydrolithon* spp. and *Titanoderma* spp. [see review by 40]. Additionally, only one study has examined the effects of climate change on the preferences of corals to settle on specific CCA species. Doropoulos et al. [37] found that *Acropora millepora* larvae preferred to settle on *Titanoderma* sp. under ambient conditions and that this preference was eliminated under increased pCO_2 . While this study suggests a possible change in larval preferences for CCA, it (1) does not take into account increases in seawater temperature that will occur with OA and (2) examines the response of only one coral species.

Here, we investigate the effects of OW and OA on photosynthetic efficiency, measured as maximum quantum yield (F_v/F_m), and net calcification rate of two species of CCA (*Hydrolithon boergeresii* and *Paragoniolithon solubile*) with implications for coral settlement. We also examine the effects of OA and OW on the settlement preferences of the brooding coral *Po. astreoides*, and two broadcast spawning corals, *A. palmata* and *A. cervicornis*, for the CCA species *H. boergeresii* and *Pa. solubile*.

2. Methods

Three experiments were performed to examine the effects of OA and OW on two species of Caribbean CCA, *H. boergeresii* and *Pa. solubile*, and the subsequent impact on coral larval settlement (Table 1). In all experiments, CCA were exposed to either ‘present-day’ seawater conditions ($\text{pH}_{\text{NBS}} \sim 8.1$ – 8.3 , temperature ~ 27 – 28 °C) or ‘future’ seawater conditions predicted to occur near the year 2100 according to the ‘business-as-usual’ RCP 8.5 IPCC emissions scenario ($\text{pH}_{\text{NBS}} \sim 7.9$ – 8.0 , temperature ~ 29 – 31 °C, Table 2). Experiments spanned multiple exposure times, ranging from 7 to 99 days, and geographic regions (Belize or Florida). All CCA were examined for treatment effects on maximum quantum yield (F_v/F_m) and attractiveness to coral settlers. CCA exposed to long-term conditioning were also evaluated for effects

Table 1

Summary of experiments including CCA collection information.

	Exp. 1: Short-term Belize	Exp 2: Short-term Florida	Exp. 3: Long-term Florida
Conditioning time	7 days	21 and 8 days	99 days
Location	Belize	Florida	Florida
CCA collection site	CBC Reef Flat	Big Pine Ledges and Wonderland Reef	Big Pine Ledges
CCA collection dates	22–27 June 2017	27 April and 15 May 2017	14 December 2017
Conditioning dates	10–17 Aug 2017	3–24 or 16–24 May 2017	5 Feb - 15 May 2018
Coral species for settlement assays	<i>A. palmata</i> , <i>A. cervicornis</i>	<i>Po. astreoides</i>	<i>Po. astreoides</i>
Type of settlement assay	choice	choice and no-choice	no-choice
CCA response metrics	F_v/F_m	F_v/F_m	F_v/F_m , calcification

on net calcification rate.

2.1. CCA collection

The CCA species *H. boergeresii* and *Pa. solubile* were selected for this study based on previously demonstrated differences in Caribbean coral settlement preferences [10].

In Belize, CCA were collected at 1 m depth along the reef flat directly adjacent to the field station at Carrie Bow Cay (CBC; $16^\circ 48.18'N$, $88^\circ 04.93'W$) in June 2017 (Table 1). After collection, excess coral rubble was trimmed from beneath the CCA using diagonal cutters, CCA fragments were identified based on morphological characteristics and a subset of each species was sampled for molecular analysis (see paragraph below ‘CCA identification’). The remaining algal material was placed in two mesh collecting bags and hung in seawater beneath the CBC dock from June 28 until August 9, 2017 for storage until use in experiments coinciding with the August spawning of acroporid corals. CCA were removed from the dock and placed in flowing seawater for 24 h prior to placement in experimental aquaria.

In Florida, CCA were collected at 5–10 m depth on 27 April 2017 (21-day conditioning experiment) from Big Pine Ledges ($24^\circ 33.213'N$, $81^\circ 22.665'W$) and Wonderland Reef ($24^\circ 33.62'N$, $81^\circ 30.08'W$) in the lower Florida Keys (Table 1). These two sites are both offshore and similar environments. On 15 May 2017 (8-day conditioning experiment) and 14 December 2017 (99-day conditioning experiment), CCA were collected from Big Pine Ledges (Table 1). CCA were transported to Mote’s Elizabeth Moore International Center for Coral Reef Research and Restoration (IC2R3) on Summerland Key where they were maintained in running seawater until they were transported in a cooler filled with seawater (within 24 h) to the Smithsonian Marine Station where they were placed in recirculating seawater aquaria until use in experiments. CCA were trimmed and sampled as above prior to use.

2.2. CCA identification

CCA were initially identified based on morphological, anatomical and reproductive characters that were evaluated using a stereo microscope. Thallus surface characteristics, margin morphology, and shape and dimension of reproductive structures (conceptacles) were some of the diagnostics used for identification [9]. While CCA taxonomy is currently evolving, we use the names *Hydrolithon boergeresii* and *Paragoniolithon solubile* here to be consistent with previous literature using the same CCA populations [10,39,41]. Regarding the use of the genus *Paragoniolithon*, we are aware that this genus is currently in synonymy with *Neogoniolithon* [42]; nevertheless, we chose to use this nomenclature sensu Adey et al. [43] because the morphological and anatomical characters of the specimens used here conform with the original description of this genus. Voucher sequences of the photosystem II

Table 2Carbonate chemistry parameters (means $\pm$ SE).

Experiment 1 (Short-term Belize; $n = 20$ seawater measurements / aquarium for salinity, temperature and pH; $n = 1$ seawater measurement / aquarium for TA)							
Scenario	Salinity	Temperature ($^{\circ}$ C)	pH	pCO ₂ (μ atm)	DIC (μ mol kg ⁻¹ SW)	Ω aragonite	TA (μ mol kg ⁻¹ SW)
Present $n = 6$	36.2 ($\pm$ 0.3)	28.3 ($\pm$ 0.2)	8.13 ($\pm$ 0.01)	402.5 ($\pm$ 8.7)	1760.7 ($\pm$ 4.2)	3.1 ($\pm$ 0.04)	2042.4 ($\pm$ 33.3)
Future $n = 6$	36.7 ($\pm$ 0.4)	30.5 ($\pm$ 0.4)	7.89 ($\pm$ 0.01)	805.6 ($\pm$ 24.5)	1890.2 ($\pm$ 7.1)	2.1 ($\pm$ 0.08)	2070.3 ($\pm$ 19.1)
Experiment 2 (Short-term Florida; $n = 16$ seawater measurements / aquarium for salinity, temperature and pH; $n = 1$ seawater measurement / aquarium for TA)							
Scenario	Salinity	Temperature ($^{\circ}$ C)	pH	pCO ₂ (μ atm)	DIC (μ mol kg ⁻¹ SW)	Ω aragonite	TA (μ mol kg ⁻¹ SW)
Present $n = 6$	33.5 ($\pm$ 0.3)	26.9 ($\pm$ 0.1)	8.30 ($\pm$ 0.03)	269.0 ($\pm$ 29.0)	1627.6 ($\pm$ 22.2)	3.3 ($\pm$ 0.2)	1952.0 ($\pm$ 27.8)
Future $n = 6$	34.1 ($\pm$ 0.5)	28.8 ($\pm$ 0.2)	7.95 ($\pm$ 0.02)	729.1 ($\pm$ 42.3)	1871.6 ($\pm$ 11.2)	2.0 ($\pm$ 0.08)	2047.5 ($\pm$ 37.4)
Experiment 3 (Long-term Florida; $n = 54$ seawater measurements / aquarium for salinity, temperature and pH; $n = 12$ seawater measurements / aquarium for TA)							
Scenario	Salinity	Temperature ($^{\circ}$ C)	pH	pCO ₂ (μ atm)	DIC (μ mol kg ⁻¹ SW)	Ω aragonite	TA (μ mol kg ⁻¹ SW)
Present $n = 6$	36.5 ($\pm$ 0.3)	27.0 ($\pm$ 0.3)	8.13 ($\pm$ 0.02)	463.1 ($\pm$ 32.9)	2027.1 ($\pm$ 50.9)	3.4 ($\pm$ 0.16)	2328.3 ($\pm$ 52.2)
Future $n = 6$	37.0 ($\pm$ 0.3)	29.9 ($\pm$ 0.3)	7.88 ($\pm$ 0.05)	1011.7 ($\pm$ 206.4)	2207.1 ($\pm$ 76.8)	2.5 ($\pm$ 0.24)	2403.7 ($\pm$ 73.0)

protein D1 gene (*psbA*) have been deposited in GenBank under accession numbers (PQ731942-PQ731950).

2.2.1. Sequencing methods

CCA surfaces were scraped with a clean razor blade into a sterile microcentrifuge tube and stored in RNAlater (Invitrogen) at -20° C. DNA extraction was performed using the Qiagen PowerSoil DNA Extraction Kit following the manufacturer's protocol. PCR amplification of the Photosystem II protein D1 gene (*psbA*) was achieved using the primers *psbA*-F1 and *psbA*-R1 [44] producing amplicons of 957bp. Samples were bi-directionally sequenced at Eurofins Genomics LLC (Louisville, Kentucky, U.S.), and Geneious Prime (version 2021.1.1, Biomatters, Auckland, New Zealand) was used to check and correct chromatograms before generating consensus sequences.

2.3. Experimental aquarium systems

Seawater carbonate chemistry for the present-day and future treatments differed across the three experiments due to variation in (1) ambient field conditions at the time of CCA collection and (2) aquarium conditions (detailed for individual experiments in sections below). Treatment offsets for pCO₂ varied from 403.1–548.6 μ atm, while offsets for temperature and Ω aragonite varied from 1.9–2.9 $^{\circ}$ C and 0.9–1.3, respectively. Due to this variation, we are only able to draw broad comparisons in responses to present-day and future seawater scenarios across coral species.

2.3.1. Belize

Crustose coralline algae and coral larvae were exposed to one of two ocean chemistry treatments: (1) present-day conditions (pH_{NBS}: 8.15; temperature 28.5 $^{\circ}$ C) or (2) future conditions (pH_{NBS}: 7.85; temperature 30.5 $^{\circ}$ C). Present-day aquaria ($n = 6$) replicated the current seawater state at Carrie Bow Cay during the time of CCA and coral larval collection, while future aquaria ($n = 6$) replicated oceanic conditions projected for the year 2100 (RCP8.5 scenario projections). Twelve independent aquaria (18.9 L mesocosms) were established indoors under ambient lighting, and randomly assigned to receive one of the two seawater treatments. All aquaria were initially filled with clean seawater from the flow-through seawater system, which has an intake at 1 m depth in the backreef lagoon adjacent to CBC. Continuous aeration was provided by a series of air-pump driven diffuser discs placed within each mesocosm. A single 50 W heater was placed in each aquarium replicating future oceanic conditions and set to impose an offset (2 $^{\circ}$ C). Temperature was not actively controlled within the present-day aquaria. Carbonate chemistry within the future aquaria was adjusted by the slow-dosing of 100 % CO₂ gas using an automated pH stat system (Aqua-Medic). pH was actively monitored within each future aquarium using independent lab-grade, double junction pH probes (Neptune Systems),

and gas flow was controlled by independent magnetic solenoids. Carbonate chemistry was not actively controlled within present-day aquaria.

Twice daily (AM and PM) measurements of pH, temperature and salinity were conducted for all aquaria to ensure stability and proper pH/temp offsets. pH and temperature were checked with an Orion Ross combination electrode, relative accuracy $\pm$ 0.03 units. This electrode was calibrated daily with NBS buffers. While it is understood that using this scale can lead to some error when calculating carbonate parameters [45] it is relatively small when making comparisons in manipulative experiments with pCO₂ offsets greater than 100 μ atm [46]. Salinity was monitored with a YSI Pro20 handheld meter, and deviations from 36 ppt were corrected with the addition of DI water. Water samples (50 ml) were collected at noon from each aquarium to measure total alkalinity (TA). After collection, samples were immediately filtered (0.2 μ m) and stored in the dark at 4 $^{\circ}$ C. TA was determined in the lab via open-cell, potentiometric titration using a Mettler Toledo DL15 titrator. Certified reference materials (alkalinity standards, Dickson Lab, UCSD) were used to ensure the accuracy of the alkalinity titrations (mean deviation from the CRM was near 2 %). All carbonate parameters were calculated using the program CO2SYS (with the measured parameters of pH, TA, temperature and salinity), using the carbonate dissociation constants of Mehrbach et al. [47], as refit by Dickson & Millero [48]. See Table 2 for a full description of all carbonate parameters.

2.3.2. Florida

Two separate experiments were conducted at the Smithsonian Marine Station in Fort Pierce, Florida using larger independent aquaria (37 L), each establishing present-day and future seawater chemistry treatments ($n = 6$). In May 2017, conditioning and settlement experiments were conducted with the following seawater treatments: present-day (pH_{NBS} / temperature of 8.30 and 27 $^{\circ}$ C, respectively) and future (pH_{NBS} / temperature of 7.95 and 28.8 $^{\circ}$ C, respectively). In February – May 2018, conditioning and settlement were conducted with the treatments of: present-day (pH_{NBS} / temperature of 8.13 and 27 $^{\circ}$ C, respectively) and future (pH_{NBS} / temperature of 7.88 and 29.9 $^{\circ}$ C, respectively).

All aquaria were housed indoors, and lighting was provided by 430 W ATI T5 HO light fixtures that replicated broad spectrum irradiance (PAR = 450 μ mol photons m⁻² s⁻¹). Each aquarium consisted of a closed seawater system, with water volume continuously recirculated by a 473 liters per hour (LPH) powerhead. Additional water flow in each aquarium was provided via airstone bubbling (with ambient air). Each aquarium was initially filled with filtered seawater (< 10 μ m) collected from an offshore oceanic location near Fort Pierce, FL. Temperature control was provided by separate water-jacketed heat exchangers attached to each aquarium, and was set and maintained at the designated treatment aquarium temperatures by dual-stage digital

controllers. Salinity was maintained by replenishing evaporative losses with deionized water. During the experiment, weekly water changes were conducted within each aquarium (100 % volume). CO₂ concentrations were manipulated via a coupled pH stat system (Aqua Medic), which monitored individual aquarium pH using separate electrodes and dosed low pH aquaria with 100 % gaseous CO₂ via magnetic solenoids. Morning and afternoon pH measurements within each aquarium were taken 4–5 times per week with an Orion Ross combination electrode to ensure proper calibration and setpoint of the pH stat system. Salinity was recorded simultaneously with pH and was measured with a YSI Pro20. Weekly water samples were collected to measure TA and carbonate parameters and were analyzed similar to the experiments performed in Belize. See Table 2 for a full description of all carbonate parameters. We note that the pH within the present-day seawater treatment was relatively high (~8.3) in Experiment 2 as compared to the other two experiments (~8.13). This was likely driven by photosynthetic pH cycling and pCO₂ drawdown within some individual aquaria, despite continuous aeration and weekly water changes. However, these high pH / low pCO₂ conditions are not entirely anomalous during the spring season in the Florida Keys [49].

2.4. Larval collection and settlement assays

In Belize, gametes of *A. palmata* and *A. cervicornis* were collected in situ at CBC during the August 2017 spawning event (six days after the August full moon). Gamete bundles were returned immediately to the lab on CBC and allowed to break apart, after which eggs and sperm from different colonies of their respective species were cross-fertilized and resulting zygotes pooled together. Developing larvae were reared for five days in flowing seawater in buckets with 180 µm mesh bottoms until swimming and competent to settle.

For Florida experiments, *Po. astreoides* colonies were collected in May 2017 (20 colonies) and May 2018 (14 colonies) at approximately 6 m depth from Wonderland Reef (24° 33.62'N, 81° 30.08'W) in the lower Florida Keys and transported in coolers to Mote's IC2R3, Summerland Key, FL. Colonies were placed in outdoor raceways with running seawater. Larvae were collected following the methods described in Kuffner et al. [50]. Colonies were placed in separate 3 L bowls, which were tilted so that released larvae spilled over the handle into separate plastic tri-pour beakers fitted with a 180 µm mesh bottom. Water levels within each beaker remained constant so that released larvae were retained until the following morning when larvae were pooled and transported to the Smithsonian Marine Station.

2.5. Experimental set-up

2.5.1. Experiment 1: short-term Belize

Initial maximum quantum yield (F_v/F_m) was measured (methods below) on all CCA, and three individual fragments (~8 cm²) of each species of CCA were added to each of 12 experimental aquaria (6 future, 6 present-day) on 10 August 2017. After conditioning for 7 days (17 August), F_v/F_m was again recorded for each CCA fragment. Two fragments of each species were used in settlement choice assays with *A. palmata* and *A. cervicornis* larvae. For these assays, one fragment of *Pa. solubile* and one fragment of *H. boergesenii* were placed in a 5.6 cm internal diameter by 10 cm long cylindrical acrylic settlement chamber similar to those used by Ritson-Williams et al. [51] with 50 5-day old larvae (spawned 12 August) of either *A. palmata* or *A. cervicornis* such that larvae of each coral species had a choice between the two CCA fragments for settlement. Chambers were returned to the same experimental aquarium the CCA had been conditioned in for the duration of the settlement assay such that each aquarium contained two chambers, one with *A. palmata* and one with *A. cervicornis* larvae. Larvae were considered settled when they had completely attached and metamorphosed onto the substrate. The number of corals settled on the surface of each CCA fragment as well as the total number of settled

corals on the entire fragment (CCA surface + exposed rock underneath) were recorded after 48 h.

2.5.2. Experiment 2: short-term Florida

Three individual CCA fragments (~8 cm²) of each CCA species were added to experimental aquaria on 3 May 2017 for 21-day conditioning. One more fragment of each CCA species was added to experimental aquaria already containing 21-day conditioning CCA on 16 May 2017 for 8-day conditioning. Initial F_v/F_m was measured for all CCA prior to conditioning. Final F_v/F_m was recorded and settlement assays for 8- and 21-day conditioned CCA were performed on 24 May 2017. The effect of OA and OW on coral settlement was tested in individual (no-choice) and choice assays using larvae from the coral *Po. astreoides*. Four assay chambers were placed within each aquarium containing one of the following: 1) *H. boergesenii* conditioned for 21 days, 2) *Pa. solubile* conditioned for 21 days, 3) *H. boergesenii* + *Pa. solubile* conditioned for 21 days (choice assay), 4) *H. boergesenii* + *Pa. solubile* conditioned for 8 days (choice assay). Fifty *Po. astreoides* larvae were added to each chamber and the number of settled corals was recorded after 48 h as in Experiment 1.

2.5.3. Experiment 3: long-term Florida

Long-term conditioning (99-day) began on 5 February 2018. Initially, F_v/F_m was recorded for each CCA fragment and surface area of live CCA was measured using the aluminum foil method [52]. Prior to these measurements, the bottom surfaces of the CCA fragments were covered with Z-spar epoxy to minimize impacts of treatment on the carbonate rock structure beneath the CCA that is not directly related to CCA physiology [26,30,53]. Initial buoyant weight measurements were taken after the first month of conditioning to allow for an acclimation period. One *Pa. solubile* and two *H. boergesenii* fragments were added to each tank. F_v/F_m and buoyant weight measurements were taken on 3 May and 4 May 2018, respectively. Settlement assays began on 15 May 2018. No-choice settlement assays were performed by placing *Pa. solubile* and *H. boergesenii* in separate settlement chambers with 60 1-day old *Po. astreoides* larvae and returning them to the same aquarium in which the individual CCA fragments had been conditioned. The number of settled corals was recorded after 48 h as in Experiment 1.

2.6. Photosynthetic efficiency and net calcification

As a proxy for photosynthetic efficiency, maximum quantum yield (F_v/F_m) was measured for each individual CCA fragment both before addition to experimental aquaria and immediately after treatment conditioning (prior to settlement experiments) using a diving pulse amplitude modulated (PAM) fluorometer (Walz Diving PAM; LED emission maximum 650 nm). The 'F' parameter is a unitless intensity value for the emitted fluorescence from Photosystem II (PSII), and F_v and F_m refer to the variable and maximum fluorescence of PS II, respectively. The ratio of F_v to F_m is commonly used to represent the proportion of open PSII reaction centers that are capable of driving photochemistry, and is interpreted as an indicator of photosynthetic performance [54]. After two hours of dark adaptation, a single saturation pulse was administered to the upper surface of each CCA fragment with a 5.5 mm active diameter fiber optic probe. All measurements were conducted at a standardized distance (10 mm) and initial fluorescence values (F_0) were between 300 and 500 units.

Net calcification was calculated for CCA exposed to long-term (99-day) conditioning in experimental aquaria based on buoyant weight measurements taken monthly during conditioning using a Sartorius balance (accuracy = 0.001 g). At each time period, the average of two buoyant weight measurements was converted to dry weight based on the density of calcite (2.71 g cm⁻³) and normalized to fragment surface area (cm²). Net calcification rate was calculated by dividing net calcification by the number of days of conditioning.

2.7. Statistical analyses

Maximum quantum yield (F_v/F_m) and settlement data were analyzed with a split-plot ANOVA using the *rstatix* package [55] in R version 4.2.3 [56] with aquarium as a blocking factor, treatment as the main between-block factor and CCA species as the main within-block factor. Net calcification rate data were also analyzed with a split-plot ANOVA using the *rstatix* package [55] in R with aquarium as a blocking factor, treatment as the main between-block factor and CCA species and month as the within-block factors. All data were tested for normality (Shapiro-Wilk Test), homogeneity of variance (Levene's Test), equivalence of

covariance (Box's M Test) and sphericity (Mauchly's Test). If the data failed the Shapiro-Wilk Test, they were evaluated using a Q-Q Plot. Data that failed to meet these assumptions were either log transformed (calcification) or arcsine square root transformed (settlement) prior to analysis. Any data that still did not meet the necessary assumptions after transformation were analyzed using a two-way mixed ANOVA using trimmed means, which is robust to deviations from assumptions [57]. If there was a significant interaction between treatment and species, pairwise comparisons were performed using a *t*-test. Statistical significance was assigned based on a $p < 0.05$ cutoff. For maximum quantum yield and net calcification, values for fragments of the same CCA species

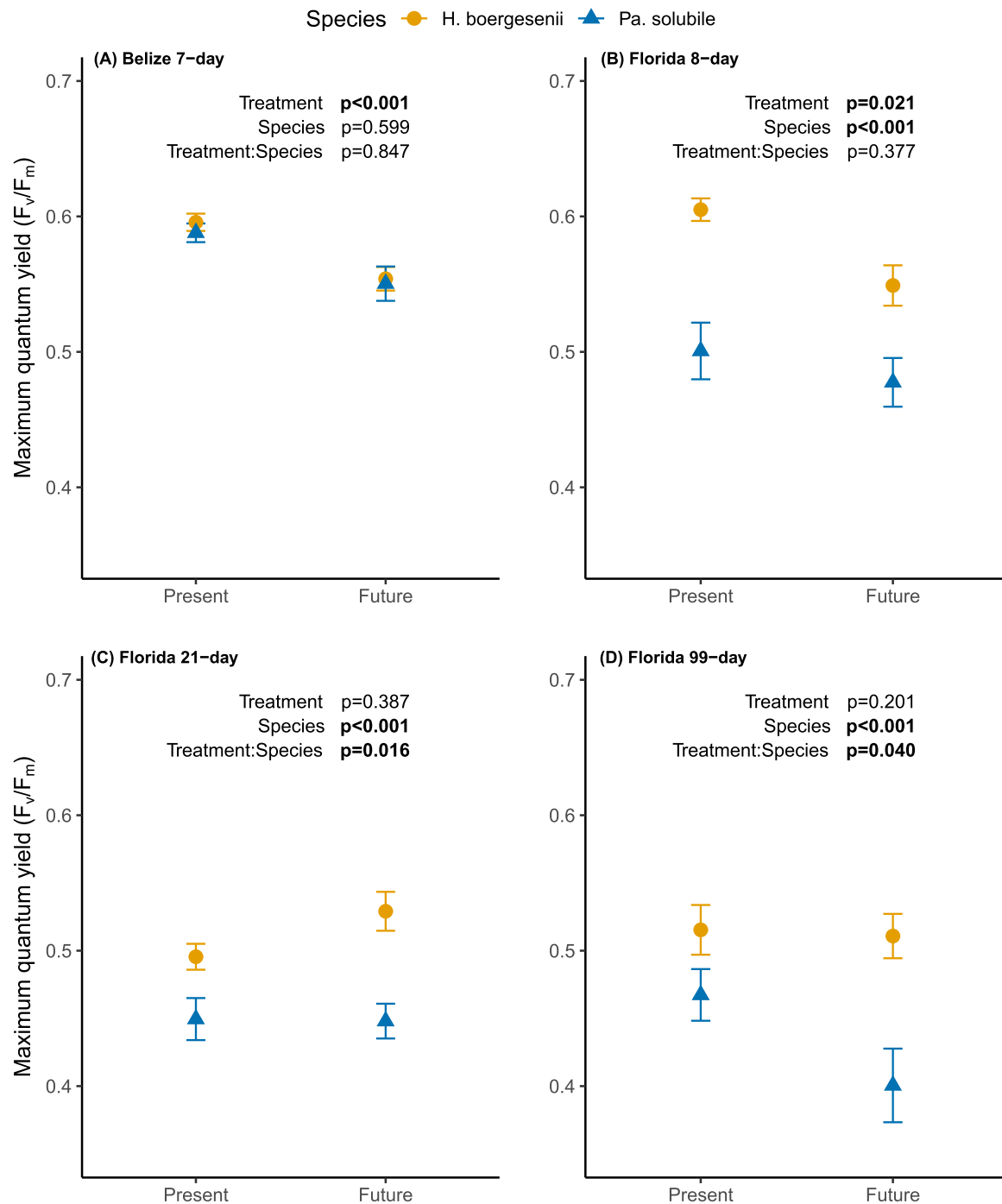


Fig. 1. Photosynthetic efficiency, measured as maximum quantum yield (F_v/F_m), of CCA *Hydrolithon boergeresii* and *Paragoniolithon solubile* after exposure to present and future seawater conditions. A) Belize 7-day exposure, B) Florida 8-day exposure, C) Florida 21-day exposure, D) Florida 99-day exposure. Symbols represent mean $\pm$ standard error. $N = 6$. Significant effects are indicated in bold.

within the same aquarium were averaged prior to analysis. All values are given as mean $\pm$ standard error.

3. Results

3.1. CCA physiology (photosynthetic efficiency and net calcification rate)

Short-term (7- or 8-day) exposure to future seawater conditions caused decreased F_v/F_m (maximum quantum yield) of CCA in Belize and in Florida, independent of CCA species (Fig. 1A and B, $p < 0.001$ and 0.021 , respectively). After longer exposure (Florida experiments only), there was an interaction between treatment and CCA species for F_v/F_m ($p = 0.016$ for 21 days, and 0.040 for 99 days). Based on pairwise comparisons, *H. boergesenii* had a higher F_v/F_m compared to *Pa. solubile* in both present-day ($p = 0.002$) and future ($p = 0.030$) conditions after 21 days (Fig. 1C, Table 3). While pairwise comparisons failed to show a significant effect of treatment within species at 21 days, there was a trend toward higher F_v/F_m in *H. boergesenii* under future conditions compared to present-day conditions ($p = 0.081$) that may be contributing to the overall interaction effect (Fig. 1C, Table 3). There was no significant effect of treatment after 99 days of exposure for *H. boergesenii* ($p = 0.856$) but there was a trend toward lower F_v/F_m in *Pa. solubile* exposed to future conditions compared to present-day conditions ($p = 0.072$, Fig. 1D, Table 3). There was no significant difference between the two CCA species in present-day conditions ($p = 0.100$), but F_v/F_m was significantly lower in *Pa. solubile* than *H. boergesenii* under future conditions ($p = 0.006$) after 99 days of conditioning (Fig. 1D, Table 3). In the long-term experiment, net calcification rate was higher in *Pa. solubile* than *H. boergesenii* ($p = 0.030$, Fig. 2), regardless of seawater condition (treatment) or time (month). There was a significant interaction between treatment and month ($p = 0.048$, Fig. 2), which was driven by a decline in calcification of both coral species under present-day conditions during months 2–3 ($p = 0.003$ and 0.037 for HB and PS respectively, Table 4). During months 1–2, net calcification rate was reduced in *H. boergesenii* exposed to future seawater conditions ($p = 0.030$, Fig. 2B, Table 4).

3.2. Coral settlement preferences

Larvae were able to settle on the surface of the CCA or onto the substrate underneath the CCA (either rock or epoxy, depending on experiment). The percentage of larvae that settled directly onto the surface of the CCA was assessed to determine settlement preferences for CCA species. Additionally, the percentage that settled on the entire CCA chip (CCA surface and rock or epoxy) was scored.

3.2.1. Choice assays

In 7- and 8-day choice assays, larvae of *A. palmata* and *Po. astreoides* settled preferentially on the surfaces of *H. boergesenii* ($25.3\% \pm 4.2$ and

Table 3

Pairwise comparisons (t-test) following significant interaction effects between exposure to present and future seawater conditions and CCA species (*Hydrolithon boergesenii* = HB, *Paragoniolithon solubile* = PS) in Florida experiments on maximum quantum yield (F_v/F_m). Significant effects are shown in bold.

	21 Days F_v/F_m		99 Days F_v/F_m	
Within Treatments				
Present	HB $\neq$ PS	$p = 0.002$	HB = PS	$p = 0.100$
Future	HB $\neq$ PS	$p = 0.030$	HB $\neq$ PS	$p = 0.006$
Within Species				
<i>H. boergesenii</i>	Present = Future	$p = 0.081$	Present = Future	$p = 0.856$
<i>Pa. solubile</i>	Present = Future	$p = 0.942$	Present = Future	$p = 0.072$

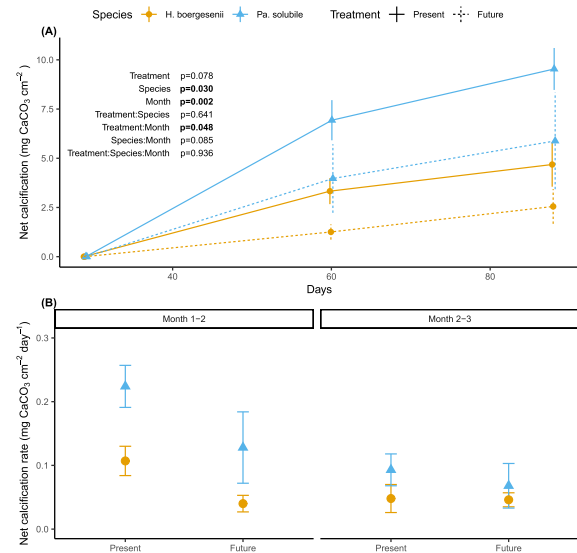


Fig. 2. (A) Net calcification over time starting after the 1-month acclimation period and (B) net calcification rate for the second and third month of CCA exposed to present and future seawater conditions. Symbols indicate mean $\pm$ standard error. $N = 6$. Significant effects are indicated in bold.

Table 4

Simple main effects test following significant interaction effects between exposure to present and future seawater conditions and exposure time (month 1–2 or month 2–3) on net calcification of *Hydrolithon boergesenii* and *Paragoniolithon solubile*.

99 Days Net Calcification		
Within Species/Treatment		
<i>H. boergesenii</i> Present	Month 1–2 = Month 2–3	$p = 0.545$
<i>H. boergesenii</i> Future	Month 1–2 $\neq$ Month 2–3	$p = 0.003$
<i>Pa. solubile</i> Present	Month 1–2 = Month 2–3	$p = 0.244$
<i>Pa. solubile</i> Future	Month 1–2 $\neq$ Month 2–3	$p = 0.037$
Within Species/Month		
<i>H. boergesenii</i> Month 1–2	Present $\neq$ Future	$p = 0.030$
<i>H. boergesenii</i> Month 2–3	Present = Future	$p = 0.934$
<i>Pa. solubile</i> Month 1–2	Present = Future	$p = 0.173$
<i>Pa. solubile</i> Month 2–3	Present = Future	$p = 0.336$

$5.5\% \pm 1.6\%$ settlement, respectively) compared to *Pa. solubile* ($3.7\% \pm 1.2\%$, $p < 0.001$ and $2.2\% \pm 0.9\%$, $p = 0.024$, respectively, Fig. 3A and C). *Acropora cervicornis* larvae did not distinguish between the two CCA species in terms of percent settlement on the CCA surface ($9.2\% \pm 3.1$ and $13.7\% \pm 4.6$ for HB and PS respectively, $p = 0.718$, Fig. 3B). The preference of *Po. astreoides* larvae for *H. boergesenii* surface was also found in choice assays after 21 days of exposure ($7.7\% \pm 1.2$ and $2.3\% \pm 0.8$ for HB and PS respectively, $p = 0.005$, Fig. 4A).

The overall percentage of *Po. astreoides* larvae that settled on CCA surfaces was low compared to total settlement within the chamber, so settlement on the entire CCA fragment (including the rock surface under the CCA) was also analyzed. When the rock surface underneath and CCA surface for each CCA fragment were analyzed together, there was no effect of CCA species on *Po. astreoides* settlement in either 8-day or 21-day choice assays ($p = 0.566$ and 0.471 , respectively, Figs. 3C and 4A), but *A. palmata* larvae maintained their preference for *H. boergesenii* ($p = 0.002$, Fig. 3A). Short-term exposure (7–21 days) to future seawater conditions did not significantly affect the settlement of any of the three coral species on CCA surfaces in choice assays. However, after 8 days of exposure to future conditions, there was a marginal increase in settlement of *Po. astreoides* on the surface of the CCA ($p = 0.066$) and decrease in settlement on the total fragment ($p = 0.068$, Fig. 3C). There were no significant interactions between CCA species and treatment in any

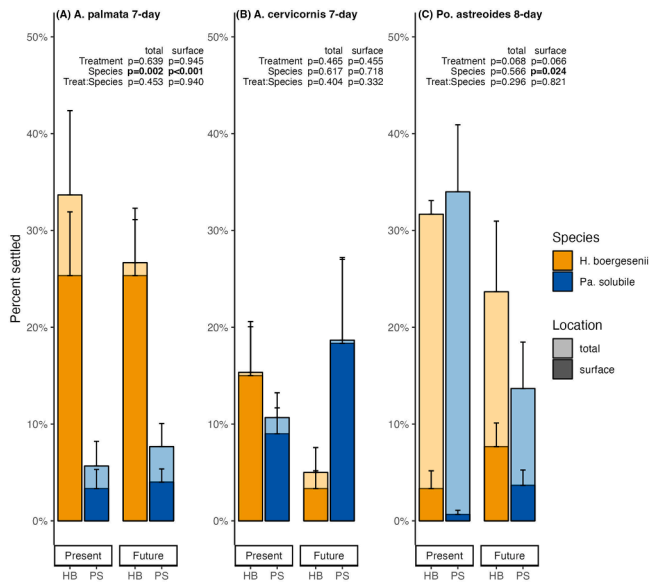


Fig. 3. Settlement of *Acropora palmata* (A), *A. cervicornis* (B) and *Porites astreoides* (C) larvae in choice assays on CCA after exposure to present and future seawater conditions for 7–8 days. Bars represent mean percentage of total larvae added to chambers that settled on the surface of the CCA (dark shade) and on the total CCA fragment including the rock underneath (light shade) $\pm$ standard error. HB = *Hydrolithon boergesenii*, PS = *Paragoniolithon solubile*. $N = 6$. Significant effects are indicated in bold.

choice assays (Figs. 3 and 4A).

3.2.2. No-choice assays

There was no significant effect of CCA species or treatment on *Po. astreoides* settlement on the CCA surface after either 21 days ($p = 0.096$ and 0.277 , respectively) or 99 days ($p = 0.168$ and 0.771 , respectively) of conditioning in no-choice assays (Figs. 4B and 5). There was a significant interaction between CCA species and treatment on *Po. astreoides* settlement on the entire fragment (CCA + rock) after 21 days of conditioning in no-choice assays ($p = 0.010$, Fig. 4B). However, pairwise comparisons failed to detect any significant differences between species within treatments or between treatments for either species. There was a trend toward an effect of treatment within *Pa. solubile* ($p = 0.066$) and between CCA species under future conditions ($p = 0.077$) but these were not significant based on a $p < 0.05$ cutoff. There was no significant effect of treatment within *H. boergesenii* ($p = 0.264$) and no difference between CCA species under control conditions ($p = 0.195$). There were no significant effects of either CCA species ($p = 0.413$) or treatment ($p = 0.773$) on *Po. astreoides* settlement on the entire fragment (CCA + epoxy) after 99 days of exposure (Fig. 5).

4. Discussion

In Florida, *H. boergesenii* displayed lower net calcification rates and higher maximum quantum yield (F_v/F_m) compared to *Pa. solubile* regardless of treatment (Figs. 1 and 2B). The difference in F_v/F_m between CCA species was also present prior to conditioning among CCA collected in Florida in spring (April/May 2017) but not winter (December 2017) (SI Table 1). Interestingly, there was no difference in F_v/F_m between the two CCA species in Belize either before or after

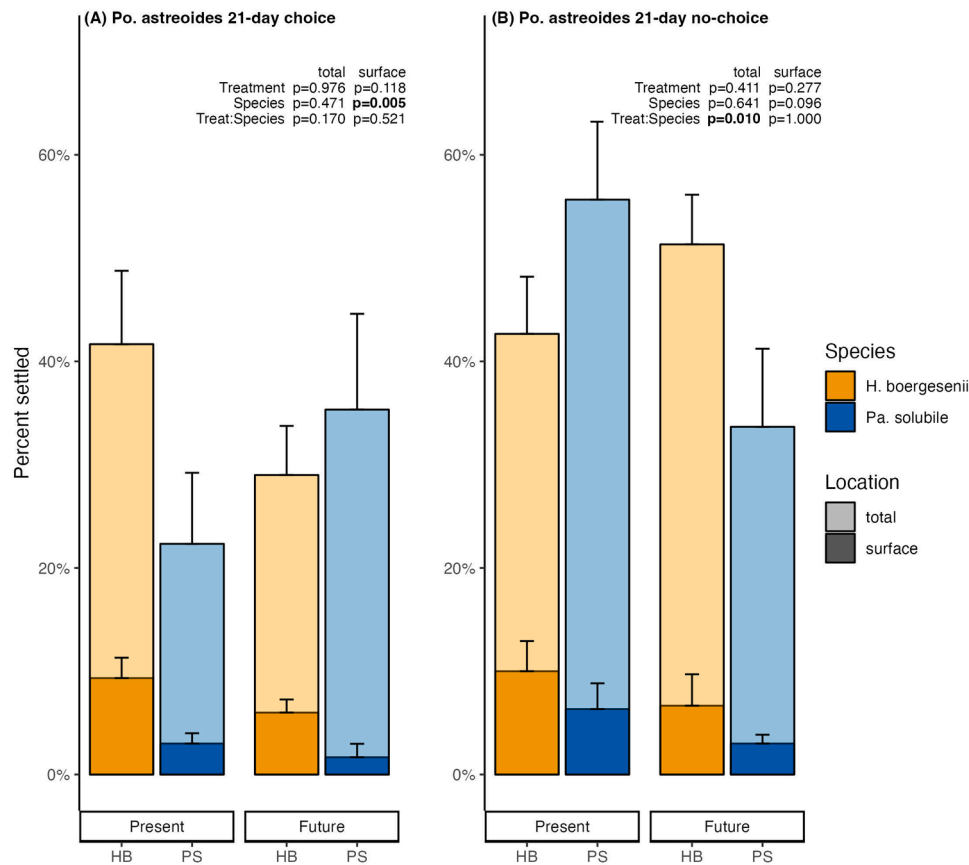


Fig. 4. Settlement of *Porites astreoides* larvae in choice (A) and no-choice (B) assays on CCA after exposure to present and future seawater conditions for 21 days. Bars represent mean percentage of total larvae added to chambers that settled on the surface of the CCA (dark shade) and on the total CCA fragment (light shade) $\pm$ standard error. HB = *Hydrolithon boergesenii*, PS = *Paragoniolithon solubile*. $N = 6$. Significant effects are indicated in bold.

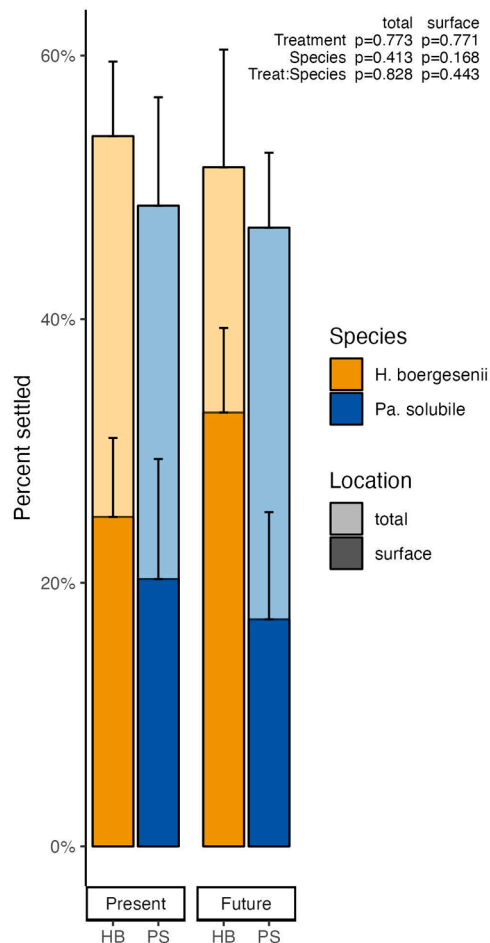


Fig. 5. Settlement of *Porites astreoides* larvae in no-choice assays on CCA after exposure to present and future seawater conditions for 99 days. Bars represent mean percentage of total larvae added to chambers that settled on the surface of the CCA (dark shade) and on the total CCA fragment (light shade) $\pm$ standard error. HB = *Hydrolithon boergesenii*, PS = *Paragoniolithon solubile*. $N = 6$.

conditioning in the 7-day experiment conducted there (SI Table 1, Fig. 1). F_v/F_m can be impacted by environmental conditions such as light intensity [58], which varied between the collection sites in Belize and Florida. CCA in Belize were collected from a very shallow (1 m depth), back reef environment and subject to high light levels, while those collected in Florida were found under ledges in deeper water (5–10 m). Although CCA collected in Florida in the winter were collected at the same sites as those collected in the spring, the lower temperature (22 °C) at time of collection may have impacted the F_v/F_m . The acclimation of these individuals to different light environments based on geography and season may have influenced their baseline F_v/F_m as well as their responses to the aquarium systems in which they were conditioned. It should be noted that while these two CCA species do co-occur in the same environments in both Belize and Florida, *Pa. solubile* is more commonly found in shallower environments than *H. boergesenii*. However, within each geographical location, the CCA individuals used in this study were collected from the same site/environment.

The CCA species examined here had similar physiological responses to expected future increases in ocean acidification and warming after relatively short conditioning periods (7–21 days) but deviated in their responses after longer conditioning (99 days). Both CCA species experienced similar declines in maximum quantum yield (F_v/F_m) in response to short-term (7–8 days) exposure to future seawater conditions in Belize and Florida (Fig. 1A and B). Over longer exposure periods, *H. boergesenii* appeared to become acclimated to future seawater conditions (Fig. 1C

and D) while *Pa. solubile* exhibited a trend toward decreasing maximum quantum yield (F_v/F_m , 14 % reduction, Fig. 1D). Decreases in maximum quantum yield (F_v/F_m) result from damage to photosystem II and therefore may be indicative of organismal stress [59]. These results indicate that both CCA species initially experience stress when exposed to future seawater conditions but that *H. boergesenii* may acclimate to these conditions over longer exposures. However, the F_v/F_m values for both species remained within the normal range for healthy CCA throughout the experiments [58,60]. This highlights the importance of experimental duration when evaluating the response of CCA to simulated environmental changes. CCA that can acclimate to OA and OW may appear susceptible to their negative effects based on shorter term experiments.

Net calcification rate was higher for *Pa. solubile* compared to *H. boergesenii* during the long-term (99-day) experiment indicating that *Pa. solubile* may be a faster calcifier (Fig. 2). It has been suggested that, among calcifying algae, slower calcifiers may be less susceptible to OA than faster calcifiers [27]. That is contrary to the results presented here in which future seawater conditions caused a decrease in net calcification rate in the slower calcifier, *H. boergesenii*, but not the faster calcifier, *Pa. solubile*. (Fig. 2). There is also evidence that some CCA are able to maintain pH at the site of calcification but that this ability differs among species, which may provide a mechanism to explain the differences in calcification responses between *Pa. solubile* and *H. boergesenii* [61]. While *H. boergesenii* experienced initial declines in calcification rate after exposure to future ocean conditions, this decline was not so extreme as to lead to net dissolution, as has been reported in some CCA species [21]. During the last month of the experiment, calcification rates declined in present-day seawater conditions for both species resulting in no significant differences in calcification rates among treatments (Fig. 2). The cause of declining calcification rates under present-day seawater conditions in the final month of the experiment is unknown and prevents any real conclusions from being drawn for that time period. Johnson et al. [53] found a similar decline in calcification rates during the latter half of a 16 week experiment. While this effect was present in both ambient and OA treatments, it occurred only under a variable pH regime and not a stable pH regime as was employed here.

Although exposure to future climate conditions caused changes to CCA physiology in some cases, this did not result in alterations to coral settlement preferences, indicating that future climate conditions will likely not impact the settlement preferences of these three coral species. There were no significant effects of treatment on coral settlement in any choice assay, although there was a marginally significant reduction in *Po. astreoides* settlement under future conditions after 8 days of exposure. After 21 days of conditioning there was a significant interaction between CCA species and treatment on the settlement of *Po. astreoides* larvae across the entire fragment (CCA + rock) in no-choice assays (Fig. 4B). Although post hoc tests failed to reveal any statistically significant differences within treatments or between treatments among species, this interaction effect was likely driven by a reduction in settlement on *Pa. solubile* fragments under future conditions compared to present-day conditions ($p = 0.066$) which resulted in a marginal difference between the CCA species ($p = 0.077$, Table 5, Fig. 4B). These results suggest that exposure to future climate conditions may cause a change in *Pa. solubile* such that these CCA become slightly less attractive to the larvae, however this effect was not present in response to CCA that had been conditioned for 21 days and subsequently used in choice assays nor those conditioned for 99 days used in no-choice assays. The general lack of a direct impact of future climate conditions on the settlement response of *Po. astreoides* has been demonstrated previously in experiments studying the effect of OA alone, not coupled with OW [34]. However, it has been suggested that OA can impact *Po. astreoides* settlement indirectly through alterations to the biofilm communities that form on settlement tiles under future conditions [34]. Similarly, *A. palmata* settlement is reduced under future OA, but whether this is caused by direct impacts to the larvae or indirect effects on the

Table 5

Pairwise comparisons (*t*-test) following significant interaction effects between exposure to present and future seawater conditions and CCA species (*Hydrolithon boergesenii* = HB, *Paragoniolithon solubile* = PS) in Florida experiments on percent settlement of *Po. astreoides* larvae.

21 Days Settlement		
Within Treatments		
Present	HB = PS	<i>p</i> = 0.195
Future	HB = PS	<i>p</i> = 0.077
Within Species		
<i>H. boergesenii</i>	Present = Future	<i>p</i> = 0.264
<i>Pa. solubile</i>	Present = Future	<i>p</i> = 0.066

settlement substrate (biofilmed tiles) is unknown [35]. The limited influence of future climate conditions on the settlement response of corals in this study may be related to the use of CCA as settlement substrata or a mediation of the impacts by combining OW with OA. Further research to tease apart these mechanisms will enhance our understanding of coral responses to future climate conditions.

In choice assays, both *A. palmata* and *Po. astreoides* demonstrated a preference for *H. boergesenii* over *Pa. solubile* in terms of settlement on the CCA surface, while *A. cervicornis* did not. These results differ somewhat from previous studies that demonstrated a preference for *H. boergesenii* over *Pa. solubile* by *A. palmata* and *A. cervicornis* but not *Po. astreoides* [10,39]. However, the majority of *Po. astreoides* larvae settled on the rock underside of the CCA fragments, not on the CCA surface, and there was no preference between CCA species when analyzing settlement on the entire CCA fragment (CCA + rock). For the acroporids, the majority of larvae settled on the CCA surface, and therefore the preference of *A. palmata* larvae for *H. boergesenii* was also present when analyzing settlement on the entire CCA fragment (CCA + rock). The preference of *A. palmata* for *H. boergesenii* over *Pa. solubile* and some other CCA species has been well documented and demonstrates a strong consistent CCA preference by this coral [9,39] while the preferences of *Po. astreoides* and *A. cervicornis* are less consistent between this pair of CCA species [39].

These results give insight into choices that larvae will make under future climate scenarios if both species of CCA are abundant and available on future reefs. In the scenario that choices may be limited in the future, the no-choice assays demonstrate that *Po. astreoides* will settle on either CCA surface when given no other choice. While not examined in this study, previous work has shown that *A. palmata* and *A. cervicornis* settle in higher numbers on *H. boergesenii* than *Pa. solubile* in no-choice assays [9,39]. The implications of these choices on the subsequent survival of the recruits is not well understood. Both *H. boergesenii* and *Pa. solubile* slough epithelial cells which could cause recruits to be dislodged from the surface of the CCA [62]. In one study, survival of *A. cervicornis* settled on a non-sloughing CCA species (*Titanoderma prototypum*) was higher than on *H. boergesenii*, but there was no difference in survival of *A. palmata* on the same pair of CCA species [39]. In general, spawning species in the Caribbean exhibit stronger preferences among CCA species compared to brooding species (i.e., *Po. astreoides*) [10]. They also exhibit stronger preferences for CCA over other non-CCA biofilmed surfaces [10]. However, brooding species often dominate natural coral recruitment in the Caribbean [63–65]. The differences in physiological responses of *Pa. solubile* and *H. boergesenii* to future climate conditions may have implications for future recruitment success of coral species that have strong preferences for *H. boergesenii* over *Pa. solubile*.

6. Conclusions

The two CCA species investigated here demonstrated differences in photosynthesis and calcification between species, and both generally exhibited negative responses to future ocean conditions. *Pa. solubile* appears to be slightly more susceptible to long-term changes in seawater

pH and temperature, while *H. boergesenii* is more resistant in terms of photosynthesis. However, calcification may be more affected in *H. boergesenii*. Furthermore, the strong settlement preferences for *H. boergesenii* over *Pa. solubile* by *A. palmata* are largely unaffected by future seawater conditions. In light of the maintenance of settlement preferences under future seawater conditions, it will be important to investigate more long-term implications of future seawater conditions on preferred CCA species.

CRedit authorship contribution statement

Jennifer M. Sneed: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Justin E. Campbell:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Audrey Looby:** Writing – review & editing, Methodology, Investigation. **Aurora Giorgi:** Writing – review & editing, Methodology, Investigation. **Valerie J. Paul:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank Nikki Fogarty, Alicia Vollmer, Megan Bock, Kelly Pitts, Zach Foltz, Molly Ashur, Zara Cowan, Jenn Joseph, and Skylar Carlson for assistance collecting CCA and coral larvae. Funding was provided by Mote Protect Our Reefs Grant program (POR 2016-7) and the Smithsonian Scholarly Studies Program. Belize Fisheries Department provided permits to conduct research at Carrie Bow Cay, Belize (permit #000009-17). In Florida, coral larvae and CCA were collected under permit #FKNMS-2017-015. **This is SMSFP contribution #1233 and CCRC contribution #1085.** This is publication #1792 from the Institute of Environment at Florida International University. We thank Julie Olson for sequencing support.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ecoecg.2024.100090.

Data availability

Relevant data and code can be found at: 10.25573/data.25290295.

References

- [1] T.A. Gardner, I.M. Cote, J.A. Gill, A. Grant, A.R. Watkinson, Long-term region-wide declines in Caribbean corals, *Science* 301 (2003) 958–960, <https://doi.org/10.1126/science.1086050>.
- [2] J.M. Pandolfi, R.H. Bradbury, E. Sala, T.P. Hughes, K.A. Bjorndal, R.G. Cooke, D. McArdle, L. McClenachan, M.J.H. Newman, G. Paredes, R.R. Warner, J.B. C. Jackson, Global trajectories of the long-term decline of coral reef ecosystems, *Science* 301 (2003) 955–958, <https://doi.org/10.1126/science.1085706>.
- [3] T.P. Hughes, J.T. Kerry, M. Alvarez-Noriega, J.G. Alvarez-Romero, K.D. Anderson, A.H. Baird, R.C. Babcock, M. Beger, D.R. Bellwood, R. Berkelmans, T.C. Bridge, I. R. Butler, M. Byrne, N.E. Cantin, S. Comeau, S.R. Connolly, G.S. Cumming, S. J. Dalton, G. Diaz-Pulido, C.M. Eakin, W.F. Figueira, J.P. Gilmour, H.B. Harrison, S. F. Heron, A.S. Hoey, J.P.A. Hobbs, M.O. Hoogenboom, E.V. Kennedy, C.Y. Kuo, J. M. Lough, R.J. Lowe, G. Liu, M.T.M. Cculloch, H.A. Malcolm, M.J. McWilliam, J. M. Pandolfi, R.J. Pears, M.S. Pratchett, V. Schoepf, T. Simpson, W.J. Skirving, B. Sommer, G. Torda, D.R. Wachenfeld, B.L. Willis, S.K. Wilson, Global warming and recurrent mass bleaching of corals, *Nature* 543 (2017) 373–377, <https://doi.org/10.1038/nature21707>.

- [4] A. Dietzel, M. Bode, S.R. Connolly, T.P. Hughes, Long-term shifts in the colony size structure of coral populations along the Great Barrier Reef, *Proc. Biol. Sci.* 287 (2020) 20201432, <https://doi.org/10.1098/rspb.2020.1432>.
- [5] C. Birkeland, Chapter 2 - Global status of coral reefs: in combination, disturbances and stressors become ratchets, in: C. Sheppard (Ed.), *World Seas: An Environmental Evaluation* (Second Edition), Academic Press, 2019, pp. 35–56.
- [6] S.A. Sandin, J.E. Smith, E.E. Demartini, E.A. Dinsdale, S.D. Donner, A. M. Friedlander, T. Konotchick, M. Malay, J.E. Maragos, D. Obura, O. Pantos, G. Paulay, M. Richie, F. Rohwer, R.E. Schroeder, S. Walsh, J.B.C. Jackson, N. Knowlton, E. Sala, Baselines and degradation of coral reefs in the northern line islands, *PLoS ONE* 3 (2008) e1548, <https://doi.org/10.1371/journal.pone.0001548>.
- [7] M.M. Littler, D.S. Littler, The nature of crustose coralline algae and their interactions on reefs, *Smithson. Contrib. Mar. Sci.* (2013) 199–212, [repository.si.edu/handle/10088/21634](https://doi.org/10.1111/j.1365-2486.2011.02404.x).
- [8] M.G. Hadfield, V.J. Paul, Natural chemical cues for settlement and metamorphosis of marine invertebrate larvae, in: J. McClintock, B. Baker (Eds.), *Marine Chemical Ecology*, CRC Press, Boca Raton, 2001, pp. 431–462.
- [9] R. Ritson-Williams, S.N. Arnold, V.J. Paul, R.S. Steneck, Larval settlement preferences of *Acropora palmata* and *Montastraea faveolata* in response to diverse red algae, *Coral Reefs* 33 (2014) 59–66, <https://doi.org/10.1007/s00338-013-1113-2>.
- [10] R. Ritson-Williams, S.N. Arnold, V.J. Paul, Patterns of larval settlement preferences and post-settlement survival for seven Caribbean corals, *Mar. Ecol. Prog. Ser.* 548 (2016) 127–138, <https://doi.org/10.3354/meps11688>.
- [11] L. Harrington, K. Fabricius, G. De'Ath, A. Negri, Recognition and selection of settlement substrata determine post-settlement survival in corals, *Ecology* 85 (2004) 3428–3437, <https://doi.org/10.1890/04-0298>.
- [12] Intergovernmental Panel on Climate Change, *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge University Press, Cambridge, 2023, <https://doi.org/10.1017/9781009157896>.
- [13] K.J. Kroeker, R.L. Kordas, R.N. Crim, G.G. Singh, Meta-analysis reveals negative yet variable effects of ocean acidification on marine organisms, *Ecol. Lett.* 13 (2010) 1419–1434, <https://doi.org/10.1111/j.1461-0248.2010.01518.x>.
- [14] S.C. Doney, V.J. Fabry, R.A. Feely, J.A. Kleypas, Ocean acidification: the other CO₂ problem, *Ann. Rev. Mar. Sci.* 1 (2009) 169–192, <https://doi.org/10.1146/annurev.marine.010908.163834>.
- [15] A.J. Andersson, F.T. Mackenzie, N.R. Bates, Life on the margin: implications of ocean acidification on Mg-calcite, high latitude and cold-water marine calcifiers, *Mar. Ecol. Prog. Ser.* 373 (2008) 265–273, <https://doi.org/10.3354/meps07639>.
- [16] A. Ordóñez, C. Doropoulos, G. Diaz-Pulido, Effects of ocean acidification on population dynamics and community structure of crustose coralline algae, *Biol. Bull.* 226 (2014) 255–268, <https://doi.org/10.1086/BBLv226n3p255>.
- [17] K.E. Fabricius, A. Klubbenschedl, L. Harrington, S. Noonan, G. De'Ath, In situ changes of tropical crustose coralline algae along carbon dioxide gradients, *Sci. Rep.* 5 (2015) 9537, <https://doi.org/10.1038/srep09537>.
- [18] I.B. Kuffner, A.J. Andersson, P.L. Jokiel, K.U.S. Rodgers, F.T. Mackenzie, Decreased abundance of crustose coralline algae due to ocean acidification, *Nat. Geosci.* 1 (2008) 114–117, <https://doi.org/10.1038/ngeo100>.
- [19] J.N. Smith, M. Mongin, A. Thompson, M.J. Jonker, G. De'Ath, K.E. Fabricius, Shifts in coralline algae, macroalgae, and coral juveniles in the Great Barrier Reef associated with present-day ocean acidification, *Glob. Chang. Biol.* 26 (2020) 2149–2160, <https://doi.org/10.1111/gcb.14985>.
- [20] V. Peña, B.P. Harvey, S. Agostini, L. Porzio, M. Milazzo, P. Horta, L. Le Gall, J. M. Hall-Spencer, Major loss of coralline algal diversity in response to ocean acidification, *Glob. Chang. Biol.* 27 (2021) 4785–4798, <https://doi.org/10.1111/gcb.15757>.
- [21] G. Diaz-Pulido, K.R.N. Anthony, D.I. Kline, S. Dove, O. Hoegh-Guldberg, Interactions between ocean acidification and warming on the mortality and dissolution of coralline algae, *J. Phycol.* 48 (2012) 32–39, <https://doi.org/10.1111/j.1529-8817.2011.01084.x>.
- [22] S. Martin, J.-P. Gattuso, Response of Mediterranean coralline algae to ocean acidification and elevated temperature, *Glob. Chang. Biol.* 15 (2009) 2089–2100, <https://doi.org/10.1111/j.1365-2486.2009.01874.x>.
- [23] A. Kato, M. Hikami, N.H. Kumagai, A. Suzuki, Y. Nojiri, K. Sakai, Negative effects of ocean acidification on two crustose coralline species using genetically homogeneous samples, *Mar. Environ. Res.* 94 (2014) 1–6, <https://doi.org/10.1016/j.marenvres.2013.10.010>.
- [24] R. Albright, Y. Takeshita, D.A. Koweek, A. Ninokawa, K. Wolfe, T. Rivlin, Y. Nebuchina, J. Young, K. Caldeira, Carbon dioxide addition to coral reef waters suppresses net community calcification, *Nature* 555 (2018) 516–519, <https://doi.org/10.1038/nature25968>.
- [25] S.L. Kram, N.N. Price, E.M. Donham, M.D. Johnson, E.L.A. Kelly, S.L. Hamilton, J. E. Smith, Variable responses of temperate calcified and fleshy macroalgae to elevated pCO₂ and warming, *ICES J. Mar. Sci.* 73 (2015) 693–703, <https://doi.org/10.1093/icesjms/fsv168>.
- [26] M.D. Johnson, R.C. Carpenter, Ocean acidification and warming decrease calcification in the crustose coralline alga *Hydrolithon onkodes* and increase susceptibility to grazing, *J. Exp. Mar. Biol. Ecol.* 434–435 (2012) 94–101, <https://doi.org/10.1016/j.jembe.2012.08.005>.
- [27] S. Comeau, P.J. Edmunds, N.B. Spindel, R.C. Carpenter, Fast coral reef calcifiers are more sensitive to ocean acidification in short-term laboratory incubations, *Limnol. Oceanogr.* 59 (2014) 1081–1091, <https://doi.org/10.4319/lo.2014.59.3.1081>.
- [28] M.D. Johnson, L.M.R. Bravo, N. Lucey, A.H. Altieri, Environmental legacy effects and acclimatization of a crustose coralline alga to ocean acidification, *Clim. Change Ecol.* 2 (2021) 100016, <https://doi.org/10.1016/j.eceochg.2021.100016>.
- [29] C.E. Cornwell, S. Comeau, T.M. Decarlo, E. Larcombe, B. Moore, K. Giltrow, F. Puerzer, Q. D'Aleixis, M.T. McCulloch, A coralline alga gains tolerance to ocean acidification over multiple generations of exposure, *Nat. Clim. Chang.* 10 (2020) 143–146, <https://doi.org/10.1038/s41558-019-0681-8>.
- [30] M.D. Johnson, V.W. Moriarty, R.C. Carpenter, Acclimatization of the crustose coralline alga *Porolithon onkodes* to variable pCO₂, *PLoS ONE* 9 (2014) e87678, <https://doi.org/10.1371/journal.pone.0087678>.
- [31] J.E. Campbell, J. Fisch, C. Langdon, V.J. Paul, Increased temperature mitigates the effects of ocean acidification in calcified green algae (*Halimeda* spp.), *Coral Reefs* 35 (2016) 357–368, <https://doi.org/10.1007/s00338-015-1377-9>.
- [32] J.-H. Kim, N. Kim, H. Moon, S. Lee, S.Y. Jeong, G. Diaz-Pulido, M.S. Edwards, J.-H. Kang, E.J. Kang, H.-J. Oh, J.-D. Hwang, I.-N. Kim, Global warming offsets the ecophysiological stress of ocean acidification on temperate crustose coralline algae, *Mar. Pollut. Bull.* 157 (2020) 111324, <https://doi.org/10.1016/j.marpolbul.2020.111324>.
- [33] K.R.N. Anthony, D.I. Kline, G. Diaz-Pulido, S. Dove, O. Hoegh-Guldberg, Ocean acidification causes bleaching and productivity loss in coral reef builders, *Proc. Natl. Acad. Sci.* 105 (2008) 17442–17446, <https://doi.org/10.1073/pnas.0804478105>.
- [34] R. Albright, C. Langdon, Ocean acidification impacts multiple early life history processes of the Caribbean coral *Porites astreoides*, *Glob. Chang. Biol.* 17 (2011) 2478–2487, <https://doi.org/10.1111/j.1365-2486.2011.02404.x>.
- [35] R. Albright, B. Mason, M. Miller, C. Langdon, Ocean acidification compromises recruitment success of the threatened Caribbean coral *Acropora palmata*, *Proc. Natl. Acad. Sci. U.S.A.* 107 (2010) 20400–20404, <https://doi.org/10.1073/pnas.1007273107>.
- [36] N.S. Webster, S. Uthicke, E.S. Botté, F. Flores, A.P. Negri, Ocean acidification reduces induction of coral settlement by crustose coralline algae, *Glob. Chang. Biol.* 19 (2013) 303–315, <https://doi.org/10.1111/gcb.12008>.
- [37] C. Doropoulos, S. Ward, G. Diaz-Pulido, O. Hoegh-Guldberg, P.J. Mumby, Ocean acidification reduces coral recruitment by disrupting intimate larval-algal settlement interactions, *Ecol. Lett.* 15 (2012) 338–346, <https://doi.org/10.1111/j.1461-0248.2012.01743.x>.
- [38] K.E. Fabricius, S.H.C. Noonan, D. Abrego, L. Harrington, G. De'Ath, Low recruitment due to altered settlement substrata as primary constraint for coral communities under ocean acidification, *Proc. R. Soc. B Biol. Sci.* 284 (2017) 20171536, <https://doi.org/10.1098/rspb.2017.1536>.
- [39] R. Ritson-Williams, V.J. Paul, S.N. Arnold, R.S. Steneck, Larval settlement preferences and post-settlement survival of the threatened Caribbean corals *Acropora palmata* and *A. cervicornis*, *Coral Reefs* 29 (2010) 71–81, <https://doi.org/10.1007/s00338-009-0555-z>.
- [40] L. Hofmann, K. Bischof, Ocean acidification effects on calcifying macroalgae, *Aquat. Biol.* 22 (2014) 261–279, <https://doi.org/10.10354/ab00581>.
- [41] J.M. Sneed, R. Ritson-Williams, V.J. Paul, Crustose coralline algal species host distinct bacterial assemblages on their surfaces, *ISME J.* 9 (2015) 2527–2536, <https://doi.org/10.1038/ismej.2015.67>.
- [42] W.J. Woelkerling, The status and disposition of Paragoniolithon (Corallinales, Rhodophyta), *Phycologia* 26 (1987) 144–148, <https://doi.org/10.2216/i0031-8884-26-1-144.1>.
- [43] W.H. Adey, R. Townsend, W. Boykins, The crustose coralline algae Rhodophyta: (Corallinales) of the Hawaiian Islands, *Smithson. Contrib. Mar. Sci.* 15 (1982) 1–74.
- [44] H.S. Yoon, J.D. Hackett, D. Bhattacharya, A single origin of the peridinin- and fucoxanthin-containing plastids in dinoflagellates through tertiary endosymbiosis, *Proc. Natl. Acad. Sci.* 99 (2002) 11724–11729, <https://doi.org/10.1073/pnas.172234799>.
- [45] U. Riebesell, V.J. Fabry, L. Hansson, J.-P. Gattuso, Guide to Best Practices for Ocean Acidification Research and Data Reporting, Luxembourg, Publications Office of the European Union, 2011, <https://doi.org/10.2777/66906>.
- [46] S.-A. Watson, K.E. Fabricius, P.L. Munday, Quantifying pCO₂ in biological ocean acidification experiments: a comparison of four methods, *PLOS ONE* 12 (2017) e0185469, <https://doi.org/10.1371/journal.pone.0185469>.
- [47] C. Mehrbach, C.H. Culberson, J.E. Hawley, R.M. Pytkowicz, Measurement of the apparent dissociation constants of carbonic acid in seawater at atmospheric pressure, *Limnol. Oceanogr.* 18 (1973) 897–907, <https://doi.org/10.4319/lo.1973.18.6.0897>.
- [48] A.G. Dickson, F.J. Millero, A comparison of the equilibrium constants for the dissociation of carbonic acid in seawater media, *Deep Sea Res. Part A Oceanogr. Res. Pap.* 34 (1987) 1733–1743, [https://doi.org/10.1016/0198-0149\(87\)90021-5](https://doi.org/10.1016/0198-0149(87)90021-5).
- [49] D.P. Manzello, I.C. Enochs, N. Melo, D.K. Gledhill, E.M. Johns, Ocean acidification refugia of the florida reef tract, *PLOS ONE* 7 (2012) e41715, <https://doi.org/10.1371/journal.pone.0041715>.
- [50] I.B. Kuffner, L.J. Walters, M.A. Becerro, V.J. Paul, R. Ritson-Williams, K. Beach, Inhibition of coral recruitment by macroalgae and cyanobacteria, *Mar. Ecol. Prog. Ser.* 323 (2006) 107–117, <https://doi.org/10.3354/meps323107>.
- [51] R. Ritson-Williams, S.N. Arnold, V.J. Paul, The impact of macroalgae and cyanobacteria on larval survival and settlement of the scleractinian corals *Acropora palmata*, *A. cervicornis* and *Pseudodiploria strigosa*, *Mar. Biol.* 167 (2020) 31, <https://doi.org/10.1007/s00227-019-3639-5>.
- [52] J.A. Marsh, Primary productivity of reef-building calcareous red algae, *Ecology* 51 (1970) 255–263, <https://doi.org/10.2307/1933661>.
- [53] M.D. Johnson, L.M. Rodriguez Bravo, S.E. O'Connor, N.F. Varley, A.H. Altieri, pH variability exacerbates effects of ocean acidification on a caribbean crustose

- coralline alga, *Front. Mar. Sci.* 6 (2019), <https://doi.org/10.3389/fmars.2019.00150>.
- [54] K. Maxwell, G.N. Johnson, Chlorophyll fluorescence—A practical guide, *J. Exp. Bot.* 51 (2000) 659–668, <https://doi.org/10.1093/jexbot/51.345.659>.
- [55] A. Kassambara, rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package version 0.6.0, (2020) <https://CRAN.R-project.org/package=rstatix>.
- [56] R Core Team, R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing, Vienna, Austria, 2023.
- [57] P. Mair, R. Wilcox, Robust statistical methods in R using the WRS2 package, *Behav. Res. Methods* 52 (2020) 464–488, <https://doi.org/10.3758/s13428-019-01246-w>.
- [58] A.A. Briggs, R.C. Carpenter, Contrasting responses of photosynthesis and photochemical efficiency to ocean acidification under different light environments in a calcifying alga, *Sci. Rep.* 9 (2019) 3986, <https://doi.org/10.1038/s41598-019-40620-8>.
- [59] E.H. Murchie, T. Lawson, Chlorophyll fluorescence analysis: a guide to good practice and understanding some new applications, *J. Exp. Bot.* 64 (2013) 3983–3998, <https://doi.org/10.1093/jxb/ert208>.
- [60] R.M. Vásquez-Elizondo, S. Enríquez, Coralline algal physiology is more adversely affected by elevated temperature than reduced pH, *Sci. Rep.* 6 (2016) 19030, <https://doi.org/10.1038/srep19030>.
- [61] C.E. Cornwall, S. Comeau, M.T. McCulloch, Coralline algae elevate pH at the site of calcification under ocean acidification, *Glob. Chang. Biol.* 23 (2017) 4245–4256, <https://doi.org/10.1111/gcb.13673>.
- [62] A. Giorgi, M. Monti, D. Maggioni, P.W. Gabrielson, R.S. Steneck, K. Kocot, J. B. Olson, DNA sequencing reveals higher taxonomic diversity of coralline algae (Corallinales and Hapalidiales, Rhodophyta) in the tropical western North Atlantic that complicates ecological studies, *Botanica Marina* (2024), <https://doi.org/10.1515/bot-2024-0021>.
- [63] A. Humanes, C. Bastidas, In situ settlement rates and early survivorship of hard corals: a good year for a Caribbean reef, *Mar. Ecol. Prog. Ser.* 539 (2015) 139–151, <https://doi.org/10.3354/meps11501>.
- [64] R.P.M. Bak, M.S. Engel, Distribution, abundance and survival of juvenile hermatypic corals (Scleractinia) and the importance of life history strategies in the parent coral community, *Mar. Biol.* 54 (1979) 341–352, <https://doi.org/10.1007/BF00395440>.
- [65] C.S. Rogers, H.C. Fitz, M. Gilnack, J. Beets, J. Hardin, Scleractinian coral recruitment patterns at Salt River submarine canyon, St. Croix, U.S. Virgin Islands, *Coral Reefs* 3 (1984) 69–76, <https://doi.org/10.1007/BF00263756>.